8.9 Momentum-Conserving Thermostats and the Dissipative Particle Dynamics (DPD) Method

In many applications, it is important to preserve the total momentum while calculating canonical thermodynamic averages, i.e. one works in an ensemble that is restricted by the additional momentum constraint. Momentum conservation plays a key role in capturing long-ranged hydrodynamic interactions in coarse-grained models of complex fluids [8]. Standard Langevin and Nosé dynamics methods fail to preserve the momentum. This is quite obvious in the case of Langevin dynamics. In Nosé dynamics we have the following differential equation for the momenta:

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = -\nabla U(\boldsymbol{q}) - \xi \boldsymbol{p}.$$

Assuming central forces, we have $\sum_{i=1}^{\mathrm{N}} \nabla_{\boldsymbol{q}_i} U = \boldsymbol{0}$, thus we obtain the following equation for the total momentum vector $\boldsymbol{p}_{\mathrm{tot}}$:

$$\sum_{i=1}^{\mathrm{N}} \frac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \frac{\mathrm{d}\boldsymbol{p}_{\mathrm{tot}}}{\mathrm{d}t} = -\xi \boldsymbol{p}_{\mathrm{tot}}.$$

If the total momentum is chosen to be zero, then it will formally remain zero along trajectories (see [78, 254] for a discussion on the affect this has on canonical sampling). However, in practice, a small numerical artifact may be introduced. Moreover, under numerical discretization, bias due to finite stepsize may be present allowing a build up of total momentum in long simulations.

An interesting potential consequence of the lack of momentum conservation is the so-called "flying ice cube" problem [172], whereby the system acquires a large total momentum but the internal momenta due to interactions among the particles becomes small.

It is desirable to construct formulations and numerical methods which exactly (i.e. up to rounding error) preserve the total momentum from step to step. One obvious approach to this problem is to simply project the momenta onto the linear momentum constraint at the end of each step (or after some number of steps). Such a projection introduces potential issues in terms of convergence order and would certainly complicate the analyses presented thus far in this book. Moreover, the optimal choice of projection is unclear and it is easy to define poor schemes (for example, modifying always the momentum of just the first particle in order to balance all the remaining components) which are likely to introduce artifacts (bias) in simulation. For this reason, there is interest in "building in" momentum conservation into the equations of motion (and indeed the integrator). Ideally this should be done in a localized and homogeneous way so that momentum is not transferred by a nonphysical mechanism between distant particles.